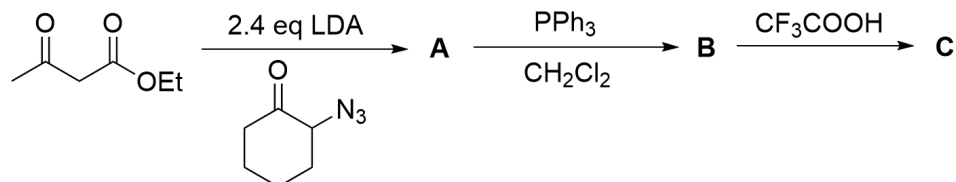


Question

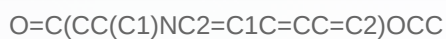
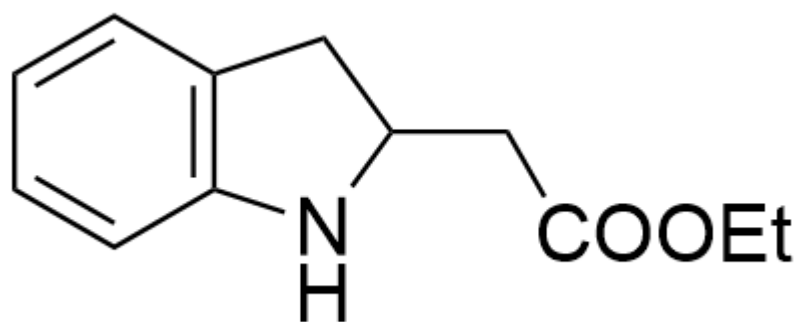


The image describes an organic tandem synthesis route. The substrate is CC(=O)CC(=O)OCC, which reacts with O=C1C(N=[N+]=[N-])CCCC1 in the presence of 2.4 equivalents of *LDA* to generate **A**. **A** reacts with *PPh*₃, *CH*₂*Cl*₂ to generate **B**. **B** reacts with *CF*₃*COOH* to generate **C**.

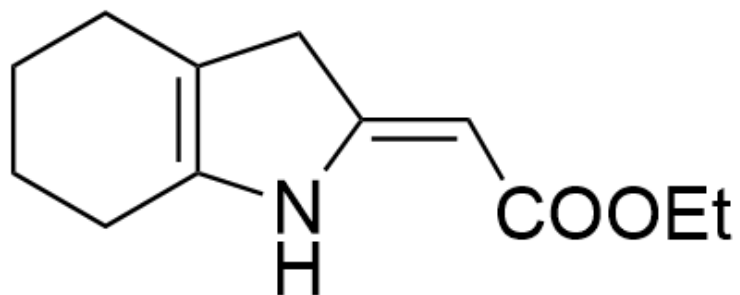
Regarding the structural formula of the final product **C** of the organic synthesis route in the above figure, the correct option is:

A. All other options are incorrect

B.

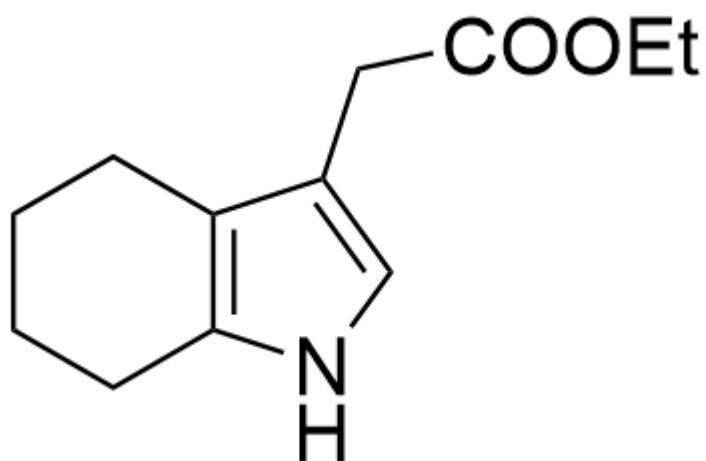


C.



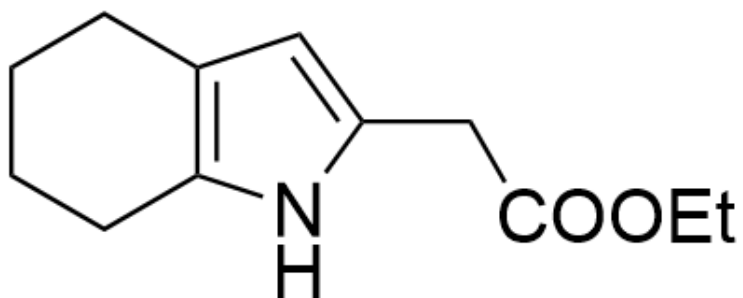
O=C(/C=C(C1)\NC2=C1CCCC2)OCC

D.



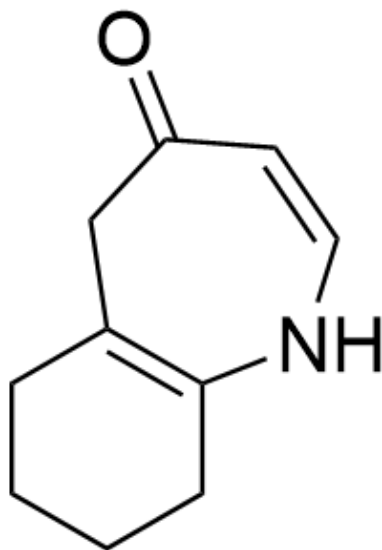
O=C(CC1=CNC2=C1CCCC2)OCC

E.



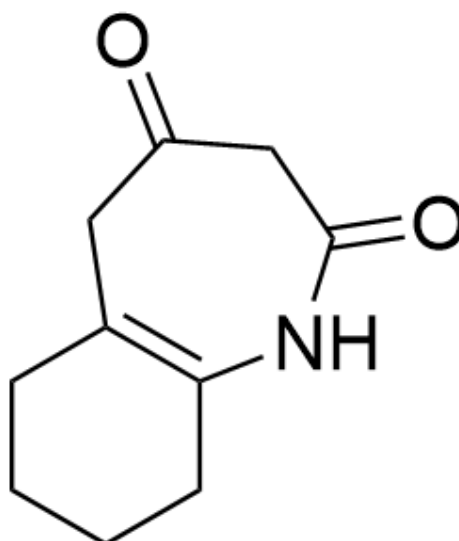
O=C(CC1=CC2=C(N1)CCCC2)OCC

F.



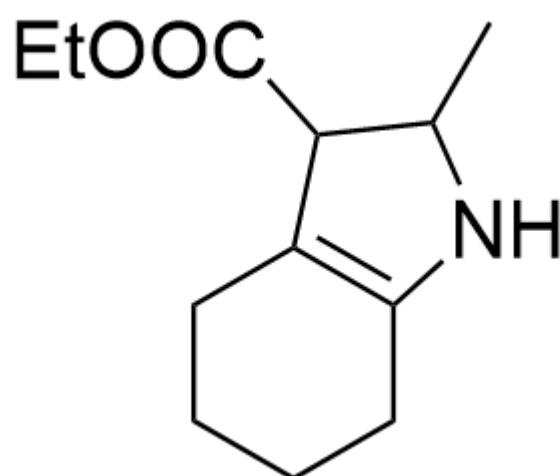
O=C(C1)C=CNC2=C1CCCC2

G.



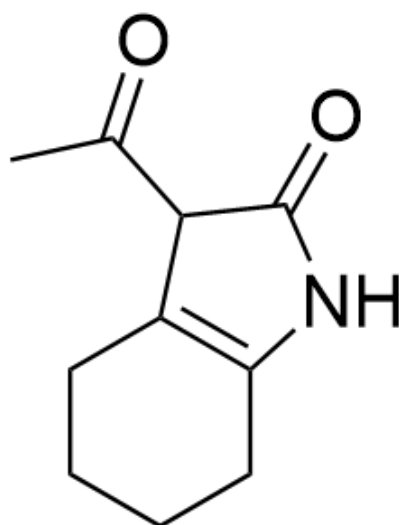
O=C(C1)CC(NC2=C1CCCC2)=O

H.



CC(C1C(OCC)=O)NC2=C1CCCC2

I.



O=C(C1C(C)=O)NC2=C1CCCC2

Answer

Correct Answer: **E**

Detailed Explanation

The first step of the reaction involves a β -dicarbonyl compound as the substrate, which can form an enolate dianion intermediate under the action of two equivalents of strong base LDA, with the structure $[\text{CH}_2^-]\text{C}([\text{CH}^-]\text{C}(\text{OCC})=\text{O})=\text{O}$.

CHECKPOINT

1 PTS

Formation of an enolate dianion intermediate under the action of two equivalents of LDA, with the structure $[\text{CH}_2^-]\text{C}([\text{CH}^-]\text{C}(\text{OCC})=\text{O})=\text{O}$

This anion can obviously undergo nucleophilic addition with another substrate containing a ketone carbonyl group. At this point, there is chemoselectivity: the methyl carbanion has a smaller degree of charge delocalization and is more nucleophilic. Under the kinetic product environment of LDA, it should preferentially nucleophilically attack over the methylene carbanion, resulting in **A** as $\text{O}=\text{C}(\text{CC}(\text{OCC})=\text{O})\text{CC}1(\text{O})\text{C}(\text{N}=[\text{N}^+]=[\text{N}^-])\text{CCCC}1$.

CHECKPOINT

1 PTS

The methyl carbanion has stronger nucleophilicity and preferentially nucleophilically attacks over the methylene carbanion

CHECKPOINT

1 PTS

A is O=C(CC(OCC)=O)CC1(O)C(N=[N+]=[N-])CCCC1

The next step is a typical Staudinger reaction. The substrate contains an azide group, which can react with triphenylphosphine to generate a phosphazene ylide. The phosphazene ylide can react with a carbonyl group. Here, chemoselectivity is also involved. The ketone carbonyl group is more electrophilic than the ester carbonyl group, and the intramolecular formation of a seven-membered ring is more difficult than the formation of a five-membered ring. Therefore, the phosphazene ylide attacks the intramolecular ketone carbonyl group, eliminating one molecule of water to form an exocyclic double bond. The structure of **B** should be OC12C(N/C(C2)=C\C(OCC)=O)CCCC1.

CHECKPOINT

1 PTS

The azide group can react with triphenylphosphine to generate a phosphazene ylide

CHECKPOINT

1 PTS

The ketone carbonyl group is more electrophilic than the ester carbonyl group

CHECKPOINT

1 PTS

Intramolecular formation of a seven-membered ring is more difficult than the formation of a five-membered ring

CHECKPOINT

1 PTS

The structure of **B** should be OC12C(N/C(C2)=C\C(OCC)=O)CCCC1

In the final step, trifluoroacetic acid is added, creating a strongly acidic environment. The intramolecular tertiary alcohol undergoes elimination, preferentially forming a fused poly-substituted double bond based on selectivity. At this point, an aza-five-membered ring olefin structure can be observed within the molecule. The acidic environment can aromatize it into pyrrole. Therefore, the final product **C** has the structure O=C(CC1=CC2=C(N1)CCCC2)OCC, and only option E is correct.

CHECKPOINT

1 PTS

Alcohol elimination preferentially forms a fused poly-substituted double bond

CHECKPOINT

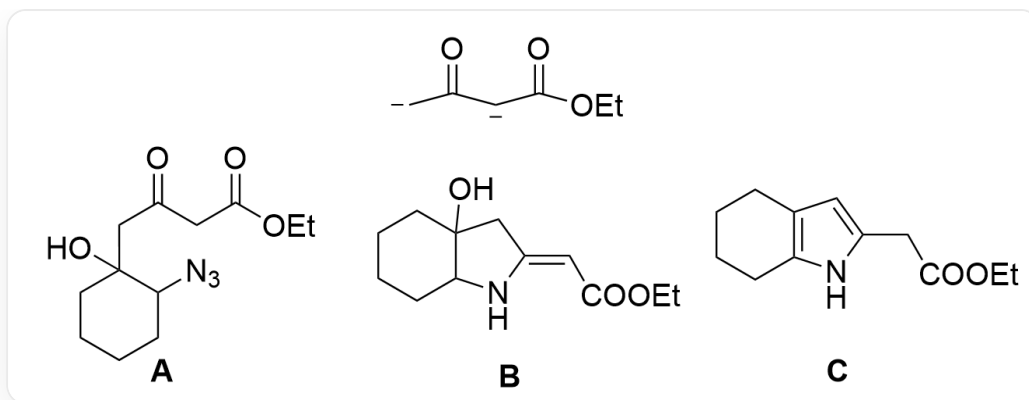
1 PTS

The acidic environment can aromatize it into pyrrole

CHECKPOINT

2 PTS

The structure of **C** is O=C(CC1=CC2=C(N1)CCCC2)OCC



This figure shows the structures of the intermediates and unknown species involved in this problem. At the top of the figure is the enolate dianion intermediate, with the structure [CH2-]C([CH-]C(OCC)=O)=O; at the bottom of the figure are the structures of the unknown species **A**, **B**, **C**, with SMILES O=C(CC(OCC)=O)CC1(O)C(N=[N+]=[N-])CCCC1; OC12C(N/C(C2)=C\C(OCC)=O)CCCC1; O=C(CC1=CC2=C(N1)CCCC2)OCC.